

A Hierarchical Framework for Inverse Problems in Biological Cybernetics: Opportunities and Limitations

Julio R. Banga¹ and Alejandro F. Villaverde^{2,3}

¹Computational Biology Lab, MBG-CSIC, Pontevedra, Spain, j.r.banga@csic.es

²Universidade de Vigo, Department of Systems Engineering and Control, Vigo, 36310, Galicia, Spain

³CITMAga, Galician Centre for Mathematical Research and Technology, Santiago de Compostela, 15782, Galicia, Spain, afvillaverde@uvigo.gal

Abstract—We examine the problem of reconstructing biological mechanisms from experimental data by means of mathematical models, a class of problems collectively known as *inverse problems*. We argue that this problem admits a natural hierarchy of increasing ambition and complexity: from estimating parameters of a known model, to discovering model structure from data, to inferring the optimality principles that govern system behavior, and finally to characterizing the strategic objectives of multiple competing biological agents. This paper examines the mathematical foundations, practical utility, and fundamental limitations of each level of this hierarchy, drawing on arguments from both proponents and critics of the framework. We assess the evidential basis for applying optimality principles to biological systems, contributing to the long-standing debate on this topic, and discuss the relationship between this hierarchy and analogous problems in modern machine learning and artificial intelligence. Our conclusion is that the lower levels rest on quite mature theoretical and computational foundations, while the upper levels represent a compelling but substantially more speculative research frontier whose biological validity must be established empirically rather than assumed on theoretical grounds. Throughout, we argue that the appropriate standard for optimality-based modeling in biology is not proof of correctness, a standard that physics itself rarely meets at the frontier, but rather demonstrated usefulness: the generation of falsifiable hypotheses, the prediction of experimental outcomes, and the organization of otherwise disconnected observations into coherent mechanistic accounts.

Index Terms—System identification, Inverse problems, Modeling, Optimal control, Differential Games.

I. INTRODUCTION

The formulation of biological understanding as an inverse problem, i.e. the recovery of hidden mechanisms from observable data, has a long and productive history in computational systems biology [10], [15], [18]. At its most elementary level, this means estimating the kinetic parameters of a known reaction network. At its most ambitious, it means inferring the evolutionary objectives that explain why a biological system behaves the way it does, or characterizing the strategic logic underlying the interactions of competing biological agents.

This paper presents and examines a four-level hierarchy of such inverse problems, assessing each level from multiple perspectives: the mathematical foundations that underpin it, the biological evidence that motivates it, and the serious objections that must be confronted before it can be applied with confidence. The four levels are:

- 1) **Parametric identification (model calibration):** Given a known model structure, estimate the parameter vector from experimental time-series data.
- 2) **Model discovery:** Simultaneously infer the model structure and its parameters from data.
- 3) **Inverse optimal control (IOC):** Infer the optimality principle (the objective function) that a biological system appears to be optimizing, given observations of its trajectories.
- 4) **Inverse differential games (IDG):** Recover the individual objectives of multiple competing biological agents from observations of their joint behavior.

Each level inherits the computational and identifiability challenges of all previous levels and adds qualitatively new ones. The lower levels (1 and 2) are relatively well-established. The upper levels (3 and 4) are subjects of active debate: they rest on empirically testable but not yet universally accepted assumptions about the nature of biological organization, and they face identifiability challenges for which complete theoretical solutions do not yet exist.

The structure of this paper reflects this asymmetry. Sections II and III treat the established foundations of the hierarchy. Sections IV and V engage more directly with the controversy, presenting supporting and critical perspectives side by side. Section VII addresses the broader question of whether optimality principles are an appropriate modeling paradigm for biology at all. Section VIII draws parallels with the analogous hierarchy in machine learning and artificial intelligence. Section X

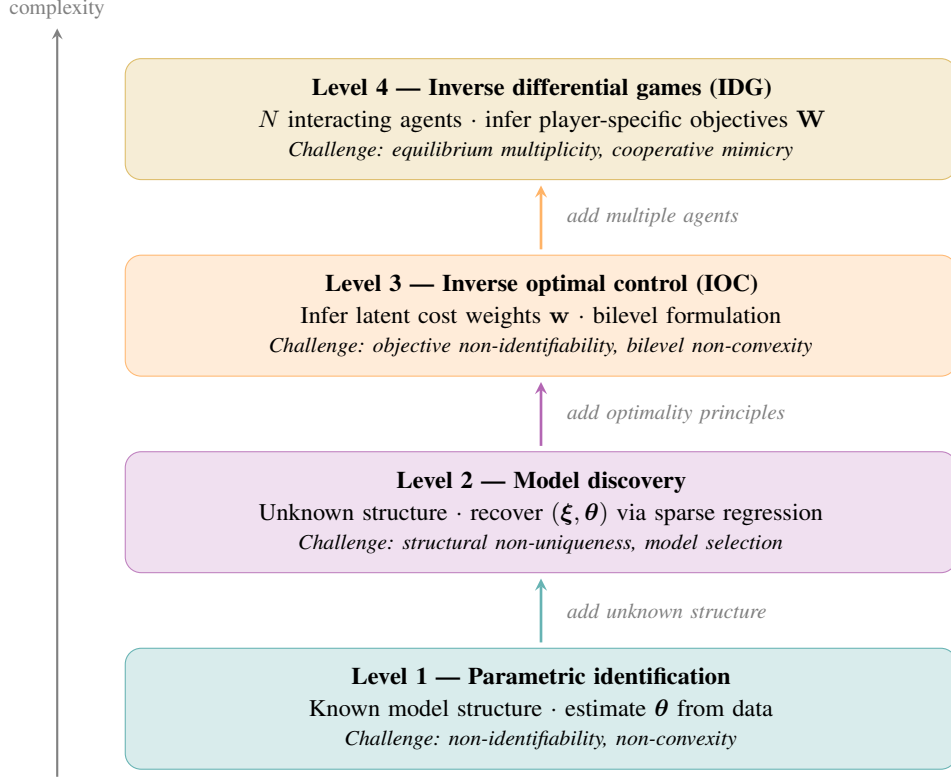


Fig. 1. Hierarchical framework for inverse problems in systems biology. Each level inherits all challenges from the levels below and introduces new sources of ill-posedness and computational difficulty.

summarizes the main conclusions and outlines a research agenda.

II. LEVEL 1: PARAMETRIC IDENTIFICATION

A. Mathematical foundations

Parametric identification is arguably the most classical form of biological inverse problem [59]. Given a state-space model:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}), \quad \mathbf{y}(t) = \mathbf{g}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}) \quad (1)$$

and experimental observations $\tilde{y}_{j,k,i}$ collected across conditions k , observables j , and time points t_i , the task is to find the parameter vector $\hat{\boldsymbol{\theta}}$ minimizing the weighted least-squares objective:

$$J(\boldsymbol{\theta}) = \sum_{k,j,i} \omega_{j,k,i} (\tilde{y}_{j,k,i} - y_{j,k,i}(\boldsymbol{\theta}))^2 \quad (2)$$

subject to dynamic constraints, box constraints on parameters, and any additional path or point constraints arising from the biology.

The key theoretical challenges are well understood. *Structural identifiability* asks whether the parameters can in principle be recovered from perfect, noise-free data

[5]; it is formally equivalent to the observability of an augmented state-space system obtained by appending $\dot{\boldsymbol{\theta}} = \mathbf{0}$ to the dynamics [56]. A parameter is structurally identifiable if and only if the corresponding component of the augmented state is observable, which can be assessed via the rank of the nonlinear observability matrix using a differential geometry approach [22], [58]. Other approaches are based on differential algebra [24], [33], generating series [32], or sensitivity matrices [57], among others. *Practical identifiability* asks whether the parameters can be reliably estimated given finite, noisy data, which depends on the conditioning of the Fisher Information Matrix and the design of the experiments [7], [30]. Common techniques for its analysis include the profile likelihood [48], Monte Carlo simulations [38], and Bayesian approaches [23].

B. Assessment

Supporting view

Level 1 is the most mature component of the hierarchy. The theoretical framework (including structural and practical identifiability analysis, plus robust optimization methods and uncertainty

quantification) is relatively well developed and has been validated in numerous biological applications [10], [16], [31], [47], [60]. The equivalence between structural identifiability and observability provides a principled diagnostic for ill-posedness before any data are collected. Practical tools for assessing and improving parameter identifiability are routinely available [49].

Critical view

Even at Level 1, the non-convexity of the optimization landscape routinely traps gradient-based methods in local minima [8]. For large models with hundreds of parameters, global optimization remains computationally expensive. Furthermore, identifiability analysis is still a challenging problem; even with recent methodological advances, the computational effort demanded by approaches both structural [49] and practical [30], [61] may be very high or even impractical for large models. Many methods for analysing structural identifiability (differential algebra, Lie derivatives, generating series) work well on models of moderate size but scale poorly. Likewise, determining practical identifiability by computing profile likelihoods or sampling-based confidence intervals can be costly. Parameter non-identifiability (the existence of multiple parameter vectors producing indistinguishable outputs) is the rule rather than the exception in mechanistic biological models, and cannot be resolved by additional data alone when it is structural in origin. Current challenges include finding biologically meaningful and experimentally feasible reparameterizations, and extending the applicability of current techniques beyond ODE models [61].

Synthesis

Level 1 is the appropriate starting point for any computational biology inverse problem, and its tools should be applied as a prerequisite before attempting higher levels. Identifiability analysis should be performed before parameter estimation, not after. The structural identifiability $\equiv$ observability equivalence provides a clean theoretical foundation that practitioners should exploit routinely.

III. LEVEL 2: MODEL DISCOVERY

A. Mathematical foundations

Model discovery extends Level 1 by treating the model structure itself as an unknown [12], [14], [17], [20], [43]. ***PENDING: discuss briefly the different approaches***. In sparse regression methods [13], [35], the system dynamics are represented as a sparse combination of candidate basis functions drawn from a library $\Phi = \{\phi_1, \dots, \phi_P\}$:

$$\dot{\mathbf{x}}(t) = \sum_{p=1}^P \xi_p \phi_p(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}_p) \quad (3)$$

and the joint optimization problem seeks both the sparse coefficient vector $\boldsymbol{\xi}$ (defining the model structure) and the continuous parameters $\boldsymbol{\theta}$:

$$\min_{\boldsymbol{\xi}, \boldsymbol{\theta}} J(\boldsymbol{\xi}, \boldsymbol{\theta}) + \lambda \|\boldsymbol{\xi}\|_q \quad (4)$$

where the regularization term $\lambda \|\boldsymbol{\xi}\|_q$ penalizes model complexity, encoding Occam's razor as a formal constraint. The hyperparameter λ governs the trade-off between fidelity to data and parsimony of the model.

B. Assessment

Supporting view

Model discovery addresses a genuine and practically important gap in computational biology: the assumption that the model structure is known is frequently untenable, and trial-and-error model building is slow and error-prone. Sparse regression approaches have demonstrated their ability to recover correct model structures from data in a range of biological settings [27]. The information-theoretic model selection criteria that accompany these approaches (AIC, BIC, Bayesian model selection) provide principled, quantitative tools for navigating the structure–complexity trade-off. The connection to AutoML and neural architecture search in machine learning (see Section VIII) suggests that the community is developing increasingly powerful tools for exactly this class of problem.

Critical view

Model discovery faces a new class of non-uniqueness that has no analogue at Level 1: structural indistinguishability. Two models with different active terms may produce identical input–output behavior for appropriately chosen parameters. In sparse regression methods, the joint

space of structures and parameters is non-smooth (setting $\xi_p = 0$ introduces a discontinuity), complicating both optimization and identifiability analysis. The choice of library Φ itself introduces a subjective element: the inferred structure can only be as good as the library from which it is drawn, and an incorrectly specified library will yield a misleading result. The regularization hyperparameter λ is a source of additional ambiguity: different values lead to different selected models, and there is no guarantee that the true model lies on the regularization path.

Synthesis

Level 2 represents a principled generalization of Level 1 that is increasingly tractable given modern computational tools. In the case of sparse regression methods, their main limitation is that it requires careful specification of the candidate library and rigorous validation of the selected structure against held-out data. Model selection at this level should be understood as providing statistical preferences, not uniqueness guarantees. Biological prior knowledge should be used to constrain the library and thereby reduce combinatorial non-uniqueness.

IV. LEVEL 3: INVERSE OPTIMAL CONTROL

A. Mathematical foundations

Inverse optimal control (IOC) introduces the hypothesis that observed biological trajectories are not merely solutions to differential equations, but solutions to optimization problems [2], [9]. The framework is bilevel: an inner optimal control problem models the system's "strategy," and an outer inverse problem recovers the unknown objective from data.

Inner problem (optimal control):

$$\mathbf{u}^* = \arg \min_{\mathbf{u}(t)} \int_{t_0}^{t_f} \sum_{m=1}^M w_m L_m(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}) dt \quad (5)$$

subject to the system dynamics and feasibility constraints on states and controls.

Outer problem (inverse optimization):

$$\min_{\mathbf{w}, \boldsymbol{\theta}, \boldsymbol{\xi}} J_{\text{outer}}(\mathbf{w}, \boldsymbol{\theta}, \boldsymbol{\xi}) + \lambda \mathcal{R}(\mathbf{w}, \boldsymbol{\xi}) \quad (6)$$

subject to $\mathbf{u}^*$ being the solution of the inner problem, normalization of $\mathbf{w}$, and parameter bounds.

B. The biological motivation

The central biological claim at Level 3 is that evolution has shaped certain biological systems to behave in ways that are approximately optimal with respect to some performance criterion. This is not a new claim; it underlies much of theoretical biology [3], [45], behavioral ecology [36], and systems biology [21], [29]. The question is whether it is sufficiently well-supported to justify the computational investment of the IOC framework.

Supporting view

The empirical evidence for optimality in biology spans a remarkable range of scales and systems. At the molecular level, the native state of a folded protein minimizes free energy [4]. At the pathway level, optimal control of enzyme expression in simple metabolic networks predicted the sequential pulse-like gene expression patterns subsequently confirmed experimentally [29], [62]. Flux balance analysis, which assumes cells maximize growth rate subject to stoichiometric constraints, predicts the growth phenotypes of *E. coli* across a wide range of genetic and environmental perturbations [25], [52]. At the organism level, the geometry of vascular branching, the design of compound eyes, and the mechanics of human gait are all well-described by optimality models [37], [41], [42]. The Klipp et al. example is particularly compelling: the optimality prediction was made *before* the experimental confirmation, satisfying the prospective prediction criterion that is often held up as the gold standard for scientific frameworks.

Critical view

The evidence for optimality is real but requires careful interpretation. Several objections deserve explicit treatment. First, a long list of confirmed predictions does not establish a framework's validity without a corresponding accounting of failures. The selectivity of evidence (presenting successes without the rate of failure) is precisely the problem identified by Gould and Lewontin [19] and by Shoemaker [51]. A systematic review of optimality predictions in molecular systems biology, including both confirmed and disconfirmed cases, is conspicuously absent from the literature. Second, the Flux Balance Analysis success story is more nuanced than it appears. Several authors have argued that FBA's predictive success reflects the tight geometric constraint structure of

metabolic stoichiometry more than any genuine optimization by the cell: in a stoichiometrically constrained network, most reasonable objective functions produce similar predictions in the relevant experimental regime. If this interpretation is correct, FBA tells us more about network geometry than about biological objectives. Third, and most importantly, identifying the *objective function* of a biological system requires not merely that the system behaves efficiently, but that the mapping from objectives to trajectories is invertible, which the theory of IOC shows it generally is not. Objective non-identifiability is a structural feature of the IOC problem, not a numerical artifact, and it compounds all the identifiability challenges already present at Levels 1 and 2.

C. The identifiability challenge

Objective identifiability at Level 3 faces a new class of fundamental difficulty. The mapping from cost functionals to optimal trajectories is many-to-one: scaling the cost by any positive constant leaves the optimal trajectory unchanged; different linear combinations of cost kernels may produce identical optimal behaviors if their gradients are collinear along the optimal trajectory; and whenever the optimal control lies on the boundary of the feasible control set (a constraint-active arc), the cost functional is effectively invisible to the inverse problem along that arc.

Alex, add refs and discussion here A necessary precondition for objective identifiability is that the inner optimal control system be controllable [28]: cost weights associated with uncontrollable dynamic modes are structurally non-identifiable regardless of data quality. Reachability provides a trajectory-specific complement: even a controllable system yields non-identifiable weights if the observed trajectories remain within a region of state-control space where different cost kernels produce indistinguishable optimal behaviors. This is the Level 3 analogue of practical non-identifiability at Level 1.

Cross-level compensation is an additional and qualitatively new source of non-uniqueness: a change in the inferred objective $\mathbf{w}$ may be offset by a compensating change in the model parameters θ , leaving the observed output unchanged. This degeneracy has no analogue at Levels 1 or 2.

Synthesis

The IOC framework is biologically motivated, mathematically coherent, and supported by a meaningful body of empirical evidence. However, it requires the following conditions to be in place before it can be applied with confidence: (i) a prior argument, independent of the framework itself, that the system of interest has been subject to sustained selection pressure on the relevant timescale; (ii) experimental designs rich enough to ensure that the observed trajectories cover a sufficiently large region of state-control space to render the objective identifiable; (iii) a strict separation between inference data and validation data, with the inferred objective tested against new experimental conditions not used in its derivation; and (iv) explicit acknowledgment of the structural identifiability limitations of the inferred objective, including the identification of cost components that are inherently non-identifiable given the system's controllability structure. The framework should be understood as generating testable hypotheses about biological objectives, not as proving what those objectives are.

V. LEVEL 4: INVERSE DIFFERENTIAL GAMES

A. Mathematical foundations

The inverse differential game (IDG) extends the IOC framework to multi-agent settings in which N biological agents interact strategically [11], [26], [39], [40]. Each agent n minimizes its own cost functional subject to the actions of all other agents, and the inner problem seeks a Nash equilibrium: a joint strategy $(\mathbf{u}_1^*, \dots, \mathbf{u}_N^*)$ from which no agent can unilaterally deviate to reduce its own cost.

The outer inverse problem seeks to recover the individual objective matrices $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_N]$ from observations of the joint equilibrium trajectories, together with the model structure ξ and parameters θ .

B. Assessment

Supporting view

The IDG framework addresses biological scenarios of genuine importance: microbial competition for shared nutrients, co-evolutionary arms races between hosts and pathogens, and competitive resource allocation in ecological communities. The suggested mathematical structure (non-cooperative differential games with Nash equilibrium inner solutions) is a natural extension of

the IOC framework and is well-supported by the theory of evolutionarily stable strategies in evolutionary biology [53], [54]. The corresponding inverse problem in artificial intelligence (multi-agent inverse reinforcement learning) is an active and well-funded research area, providing a rich set of computational tools and theoretical results that can be adapted to biological applications. For linear-quadratic differential games, conditions for the unique recovery of individual quadratic cost matrices from observed equilibrium trajectories have been derived under specific rank conditions on the closed-loop gains, providing at least a partial theoretical foundation (**add refs**).

Critical view

Level 4 is the most speculative component of the hierarchy. In addition to all difficulties from previous levels, it introduces: (i) the fundamental non-uniqueness of Nash equilibria, i.e. for given individual objectives, multiple self-consistent outcomes may coexist, and the data may correspond to any one of them without an independent theory of equilibrium selection; (ii) objective permutation symmetry, in which exchanging the objectives of two players with symmetric roles leaves the equilibrium unchanged; (iii) cooperative mimicry, in which the collective equilibrium behavior of N strategic agents is indistinguishable from the optimal behavior of a single aggregate agent, making the multi-agent structure undetectable from output data alone; and (iv) the non-identifiability of the number of agents N itself, which is analogous to the model selection problem at Level 2 but substantially harder. The assumption that biological agents are playing non-cooperative differential games (rather than simply responding to environmental signals through evolved regulatory programs) is a strong mechanistic claim that requires independent justification in each application. Bacteria competing for a carbon source are shaped by selection, not by strategic computation (**refs**). Conflating these might introduce a category error that the mathematical formalism does not automatically resolve. Furthermore, the data requirements for Level 4 are severe: recovering N individual objective vectors from joint trajectory data, in the presence of all the non-uniqueness sources enumerated above, requires experimental designs of extraordinary richness that are not yet feasible

in most biological systems.

Synthesis

Level 4 is best understood at present as a research agenda rather than a deployable methodology. Its mathematical foundations are sound, its biological motivation is conceptually coherent, and its connections to the machine learning literature are genuine and potentially productive. However, the theoretical conditions for reliable inference of individual agent objectives from joint trajectory data are not yet established for general nonlinear systems, and no end-to-end validation (infer objectives from data, predict new behavior, confirm experimentally) has been demonstrated in a real biological multi-agent setting. The appropriate near-term goal is to establish such demonstrations in well-controlled model systems (e.g., defined microbial co-cultures with known competitive interactions) where ground-truth objectives can be at least partially characterized independently.

VI. AN ILLUSTRATIVE CASE STUDY

****This could be a good point to place the case study: after the four levels have been defined, and before moving to more philosophical discussions. If we show here with an example that the theory more or less works, it could soften the more skeptical readers. It would also tell us how much we would need to shorten the next Sections.****

VII. THE BROADER QUESTION: ARE OPTIMALITY PRINCIPLES APPROPRIATE FOR BIOLOGY?

*****Pending: update references***** The upper two levels of the hierarchy rest on the biological claim that evolution produces systems that are, at least approximately, optimal with respect to some identifiable objective. This claim has a long and contested history, and its assessment has direct consequences for the practical utility of Levels 3 and 4. Below we provide a condensed view, more details can be found in [9].

A. Historical context and the Gould–Lewontin critique

The application of optimality principles to biology dates at least to Varro’s conjecture about honeycomb geometry [34] and received formal treatment in the mid-twentieth century through the work of Rashevsky, Rosen, and others [50]. The optimality research program in evolutionary biology reached its high-water mark in the 1970s and 1980s, when behavioral ecology and life-history theory generated a substantial body of confirmed predictions

using optimization models of animal behavior, foraging strategies, and reproductive allocation [45], [46].

The same period produced the most serious critique of the program. Gould and Lewontin’s “Spandrels of San Marco” [19] argued that the adaptationist program (the tendency to explain every biological trait as an optimal adaptation to some function) was structured to be immune to refutation. When a predicted optimum failed to match observation, the modeler could always escape by redefining the currency being optimized, relaxing the assumed constraints, or appealing to historical contingency. The framework was, in this reading, not a scientific theory but a post-hoc rationalization engine. This critique remains influential and cannot be dismissed. However, it is important to understand precisely what it established. The spandrels argument is a critique of *unconstrained* adaptationism, the tendency to freely choose both the objective and the constraints until the data are explained. It is not a demonstration that optimality principles are never applicable in biology. A substantial community of evolutionary and systems biologists defended the use of optimality reasoning while conceding that constraints must be specified independently of the data being explained [3], [45], [53], [55].

B. The proof versus usefulness distinction

A productive resolution of this long-standing debate comes from distinguishing between two questions that are frequently conflated:

- 1) Can optimality principles in biology be *proven correct* in the way that, say, the principle of least action is derived in classical mechanics from the Hamiltonian formalism?
- 2) Are optimality principles in biology *useful* in the scientific sense: do they generate novel hypotheses, motivate new experiments, and organize observations into coherent mechanistic accounts?

The answer to the first question is no, and this concession should be made explicitly. The principle of least action in physics rests on a mathematical structure (the Hamiltonian) that is exact, universal within its domain, and derivable from first principles. Nothing in biology enjoys this status. Biological optimality is always approximate, always domain-specific, and always conditional on constraints that must themselves be characterized empirically. To demand the same epistemological standard of biological optimality modeling as of Hamiltonian mechanics is to set a bar that physics itself rarely meets at the research frontier.

The answer to the second question is yes, with important qualifications. Scientific frameworks do not need to be

fully correct to be productive. Newtonian mechanics is “wrong” in the sense that it breaks down at relativistic speeds; the ideal gas law is “wrong” in the sense that real gases deviate from it; the Hardy-Weinberg principle is “wrong” in the sense that real populations are subject to selection, drift, and migration. None of these frameworks has been abandoned, because they are useful: they organize observations, generate predictions, and serve as baseline models against which more complex theories are assessed.

Optimality modeling in biology satisfies the usefulness criterion in the same sense. It generates hypotheses (e.g. about which metabolic objective a pathway is maximizing, about what constraint structure explains a regulatory pattern) that can be tested experimentally. When those predictions are confirmed (as in the gene expression pulse case), the framework has earned its place. When they fail (as they sometimes do), the failure is itself informative, pointing toward missing constraints or misspecified objectives. The appropriate attitude toward biological optimality is neither uncritical acceptance nor blanket rejection, but the same empirical scrutiny that any scientific framework deserves.

C. Conditions for responsible application

The “useful rather than correct” framing does not, however, constitute a free pass. Usefulness must be falsifiable to be scientifically meaningful. Several conditions should govern the responsible application of optimality modeling in computational biology:

- 1) **Prior biological justification.** An argument, independent of the data being modeled, that the system of interest has been subject to sustained and directional selection pressure on the relevant timescale. Systems that have evolved in stable environments over many generations under strong selection (e.g., central metabolic pathways of model organisms) are better candidates than recently diverged systems or those under relaxed selection.
- 2) **Pre-specified constraints.** The constraints of the optimization problem should be derived from independent biological knowledge (e.g. physical limitations, resource budgets, known regulatory architectures), not fitted to the data. The Gould-Lewontin critique has full force against post-hoc constraint specification.
- 3) **Prospective validation.** Inferred optimality principles should be used to make predictions about system behavior under new experimental conditions not used in the inference step. Retrospective fit to training data is insufficient.
- 4) **Explicit treatment of identifiability.** The structural identifiability of the inferred objective must be

TABLE I
FORMAL CORRESPONDENCE BETWEEN THE BIOLOGICAL INVERSE PROBLEM HIERARCHY AND MACHINE LEARNING.

Biological level	ML/AI analogue	Status in ML
Level 1: Parametric identification	Neural network training	Mature, deployed at scale
Level 2: Model discovery	AutoML, architecture search	Mature, industrially deployed
Level 3: Inverse optimal control	Inverse reinforcement learning	Active research, constrained domains
Level 4: Inverse differential games	Multi-agent IRL	Open research frontier

assessed, and non-identifiable components must be reported rather than suppressed.

- 5) **Honest reporting of failures.** Instances where optimality predictions failed should be reported alongside successes. The scientific community cannot assess the base rate of success of the framework without access to both.

These conditions are demanding. They might rule out a fraction of published optimality modeling in biology, which does not meet all of them. But they are the conditions under which optimality modeling earns its place as rigorous science rather than sophisticated curve-fitting.

VIII. CONNECTIONS TO MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

A striking feature of the four-level hierarchy is that each level has a close mathematical relative in modern machine learning and artificial intelligence, and those relatives have been deployed at scale with considerable success. This parallel deserves careful examination, as it provides both encouragement and important cautions for the biological application.

A. The formal correspondences

Table I summarizes the correspondence between the four levels of the biological inverse problem hierarchy and their machine learning analogues.

B. What the correspondence supports

Pending: add/update refsThe machine learning evidence provides genuine encouragement for the biological hierarchy. Neural network training demonstrates that high-dimensional, non-convex parameter estimation is tractable at massive scale with modern optimization methods. AutoML and neural architecture search demonstrate that simultaneous structure and parameter identification is achievable and practically useful. Inverse reinforcement learning (IRL) [1], [6], [44] demonstrates that inferring reward functions from observed behavior is a well-posed

and tractable problem in structured settings, and underlies much of the recent progress in learning from human demonstration in robotics and autonomous systems. Multi-agent IRL (***add refs***) demonstrates that the Level 4 problem is being actively worked on by a large community with real progress on benchmark problems.

Crucially, the machine learning analogy is not merely a structural observation. The computational tools developed for these problems (scalable optimization algorithms, regularization techniques for sparse recovery, variational methods for IRL, game-theoretic solution concepts for multi-agent settings) might be transferable (***or synergistic?***re-write this***) to the biological domain. The biological application is the frontier, since the computational foundations are not speculative.

C. The disanalogies that matter

However, the correspondence has significant limits, and the biological application cannot simply inherit the ML success story without confronting them.

Data regime. Neural networks are trained on datasets of millions to billions of examples. IRL in robotics typically requires thousands of demonstrated trajectories. Biological time-series experiments (even well-funded ones) produce tens to hundreds of observations per condition, from systems with hundreds to thousands of interacting variables. The favorable scaling properties of ML methods are not incidental: they are essential to why those methods work. Importing the framework without the data is, at best, a research challenge; at worst, it produces misleading results.

Stationarity. Machine learning systems typically operate in stationary environments: the data distribution does not change during training, and the model being learned has fixed structure. Biological cells are non-stationary, adaptive systems whose regulatory state changes in response to the perturbations used to identify them. This violates assumptions that underlie much of both classical system identification and modern machine learning. ***clarify

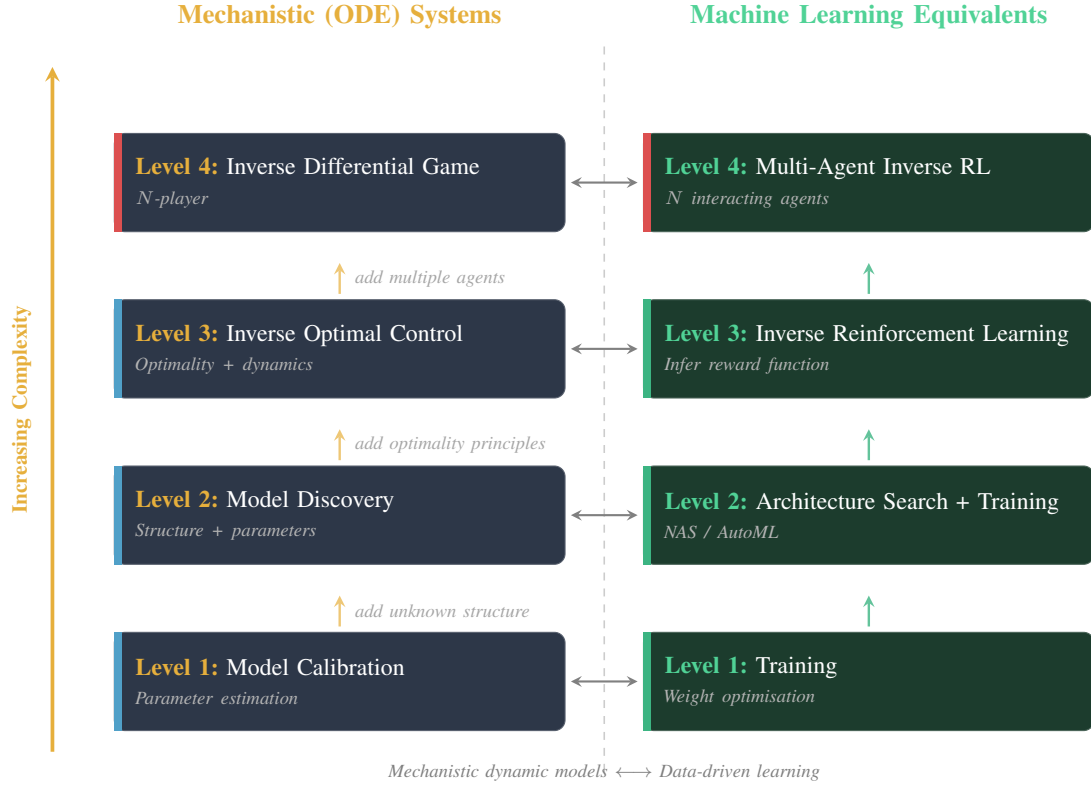


Fig. 2. The hierarchy of inverse problems in systems biology (left column) and their machine learning equivalents (right column). Each level inherits the challenges of the levels below and introduces new sources of ill-posedness and computational difficulty. The horizontal arrows indicate mathematical equivalences that this thesis exploits to develop unified methodological frameworks.

that a hybrid dynamical systems approach might be a better option, but increasing complexity etc.***

Goals of inference. In ML applications, IRL is used primarily for behavioral mimicry: the goal is to build a system that acts like the demonstrated agent, and whether the inferred reward function corresponds to the agent’s “true” objective is largely irrelevant. In biology, the goal is often mechanistic inference: the inferred optimality principle is supposed to tell us something real about the organism’s evolutionary history and regulatory architecture. This difference in epistemological goals means that the ML success story does not straightforwardly transfer. A model that mimics observed trajectories accurately but for the wrong mechanistic reasons is a useful engineering tool but a misleading scientific claim.

State of multi-agent IRL. Multi-agent IRL is not a solved problem in machine learning (***add refs***). It works well in low-dimensional, fully observable, stationary environments with small numbers of agents. It faces serious open challenges in high-dimensional, partially observable, non-stationary settings, precisely the features of real biological systems. Citing the existence of a research frontier in ML as evidence that the corresponding

biological problem is tractable might be regarded as overstating the case.

Synthesis

The ML correspondence is genuinely encouraging and practically useful: the computational tools being developed for IRL and multi-agent IRL are natural candidates for adaptation to biological inverse problems. However, the biological application faces data, stationarity, and goal-of-inference challenges that have no direct counterpart in the ML settings where these methods succeed. Closing these gaps is a concrete and tractable research agenda, but it requires explicit methodological work, not the assumption that ML success implies biological tractability.

IX. A COMPARATIVE ASSESSMENT OF THE FOUR LEVELS

Table II provides a unified comparison of the four levels across several dimensions relevant to their practical application in computational biology.

TABLE II
COMPARATIVE ASSESSMENT OF THE FOUR LEVELS OF THE BIOLOGICAL INVERSE PROBLEM HIERARCHY.

Dimension	Level 1	Level 2	Level 3	Level 4
Theoretical maturity	High	Moderate–high	Moderate	Low
Biological evidence base	Extensive	Extensive	Substantial	Limited (<i>or</i> “ <i>Partial</i> ”?)
Key non-uniqueness	Parameter non-identifiability	Structural indistinguishability	Objective non-identifiability	Equilibrium multiplicity
Key control concept	Observability	Observability (I/O equiv.)	Controllability and reachability	Joint reachability, competitive excitation
Data requirements	Moderate	Moderate–high	High	Very high
Computational burden	Moderate	High	Very high	Extreme
ML analogue maturity	High	High	Moderate	Low
Validation status	Well validated	Validated in case studies	Partial prospective validation	No end-to-end biological validation

The table makes explicit an asymmetry that runs through the entire discussion: the lower half of the hierarchy (Levels 1 and 2) is substantially better established (theoretically, empirically, and computationally) than the upper half (Levels 3 and 4). This asymmetry has important implications for how the framework should be presented and applied. Describing all four levels as components of a unified, deployable methodology overstates the case for the upper levels. A more honest characterization is that Levels 1 and 2 constitute established practice, while Levels 3 and 4 constitute a research frontier with clear motivation, coherent mathematics, and substantial open questions.

X. CONCLUSIONS

We have proposed and examined a four-level hierarchy of inverse problems in computational biology from both supporting and critical perspectives. The following conclusions emerge from this assessment.

On Levels 1 and 2. Parametric identification and model discovery rest on rather mature theoretical and computational foundations. The tools for identifiability analysis, global optimization, and model selection are quite well developed and should be part of the standard toolkit for computational biology. These levels are appropriate for immediate application, provided that identifiability

analysis is performed as a prerequisite and that model structures are validated against held-out data.

On Levels 3 and 4. Inverse optimal control and inverse differential games are biologically motivated and mathematically coherent, but they face identifiability challenges that are substantially more severe than those at Levels 1 and 2, and their biological validity depends on assumptions about evolutionary optimization that must be established empirically for each application. They should be approached as hypotheses-generating frameworks rather than inference engines, and any inferred objective or equilibrium structure should be subjected to prospective experimental validation.

On optimality principles in biology. The appropriate standard for biological optimality modeling is usefulness, not proof of correctness. Optimality principles cannot be derived from first principles in biology as they can in Hamiltonian mechanics. They can, however, generate precise and falsifiable predictions that have been confirmed in a meaningful number of cases. The conditions for responsible application (independent biological justification, pre-specified constraints, prospective validation, explicit identifiability analysis, and honest reporting of failures) are demanding but achievable. Meeting them is the difference between scientific progress and sophisticated

retrospective rationalization.

On the machine learning connection. The formal correspondence between the four biological levels and the analogous ML hierarchy (neural network training, AutoML, inverse reinforcement learning, multi-agent IRL) is genuine and practically useful. It provides computational tools, theoretical frameworks, and existence proofs that the problems are tractable in structured settings. However, the biological application faces data scarcity, non-stationarity, and mechanistic inference goals that create a substantial gap between ML success and biological applicability. Closing this gap is a concrete and well-defined research agenda.

Future directions. The most urgent priorities are: (i) end-to-end prospective validation of the Level 3 IOC pipeline in a real biological system, demonstrating that an inferred objective can predict system behavior under qualitatively new experimental conditions; (ii) development of identifiability theory for the joint $(\mathbf{w}, \xi, \theta)$ problem at Level 3; (iii) benchmark studies in defined biosystems (e.g. microbial co-culture systems) to assess the feasibility of Level 4 IDG inference; and (iv) systematic reporting of both successful and failed optimality predictions to allow the community to assess the base rate of success of the framework across biological domains.

In our view, the ladder of inverse problems offers a coherent and ambitious vision for computational biology. Climbing it responsibly requires matching the level of methodological rigor to the level of biological certainty and, at every rung, about how much of the structure above has been built and how much remains to be constructed.

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